

Project 3: The Ising Model

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1 Introduction

The Ising model is one of the most important models in statistical physics. Originally created to model the behavior of ferromagnets, the Ising model has also been applied to many other systems involving local interactions, including other magnetic orders like antiferromagnets, binary alloys, and even fluids. This project explores the results of this model, using concepts such as the correlation function and block spin transformation.

2 The Metropolis Algorithm

One way to gain insight into a physical problem is to investigate it numerically. For the Ising model, it is not realistic to compute the probabilities of all possible configurations directly, since their number grows exponentially with system size. Instead, we use a Monte Carlo simulation, which generates representative configurations of the system using random sampling.

The basic idea is to evolve the system by randomly attempting to flip individual spins and deciding whether to accept or reject each change based on the resulting change in energy. At each step, a site on the lattice is chosen at random, and the change in energy associated with flipping that spin is

computed,

$$\Delta U = U_{\text{new}} - U_{\text{old}}.$$

If the change in energy is negative or zero, the spin flip is always accepted. If the change is positive, the spin is flipped with probability

$$P = e^{-\Delta U/T}.$$

This ensures that energetically favorable moves are always accepted, while energetically unfavorable moves are still possible. As the simulation is iterated many times, the system approaches thermal equilibrium and produces configurations distributed according to the Boltzmann distribution. This process is known as the **Metropolis algorithm**.

The reason this method works is that it preserves the correct ratio of probabilities between neighboring states. Suppose two configurations differ by only a single spin flip and have energies U_1 and U_2 . According to Boltzmann statistics, in thermal equilibrium the probability of a configuration is proportional to

$$P \propto e^{-U/T}.$$

Therefore, the ratio of probabilities of the two states should satisfy

$$\frac{P_2}{P_1} = e^{-(U_2 - U_1)/T} = e^{-\Delta U/T}.$$

The Metropolis algorithm is constructed so that the probability of transitioning between these two states preserves this same ratio. If the second state has lower energy, then $\Delta U < 0$, and the transition is always accepted. If the second state has higher energy, then the transition is accepted only with probability $e^{-\Delta U/T}$. As a result, low-energy states are favored, but higher-energy states can still occur occasionally due to thermal fluctuations. This is physically important because real systems at finite temperature are never perfectly ordered. The occasional acceptance of higher-energy configurations allows the simulation to explore many possible states instead of becoming trapped in a local minimum.

Because the transition probabilities reproduce the same ratios as the Boltzmann distribution, repeated application of the algorithm causes the simulation to evolve toward the correct equilibrium statistics. After many iterations, configurations appear with frequencies proportional to their Boltzmann

probabilities, allowing thermodynamic quantities such as magnetization and energy to be estimated directly from the simulation.

Strictly speaking, the algorithm approaches the exact Boltzmann distribution only in the limit of infinitely many iterations. In practice, however, after sufficiently many updates the system reaches an approximate thermal equilibrium that is accurate enough to measure physical quantities. This also reflects the fact that real physical systems do not explore all possible microstates completely, but instead sample only a representative subset over finite timescales.

3 Problem 8.29

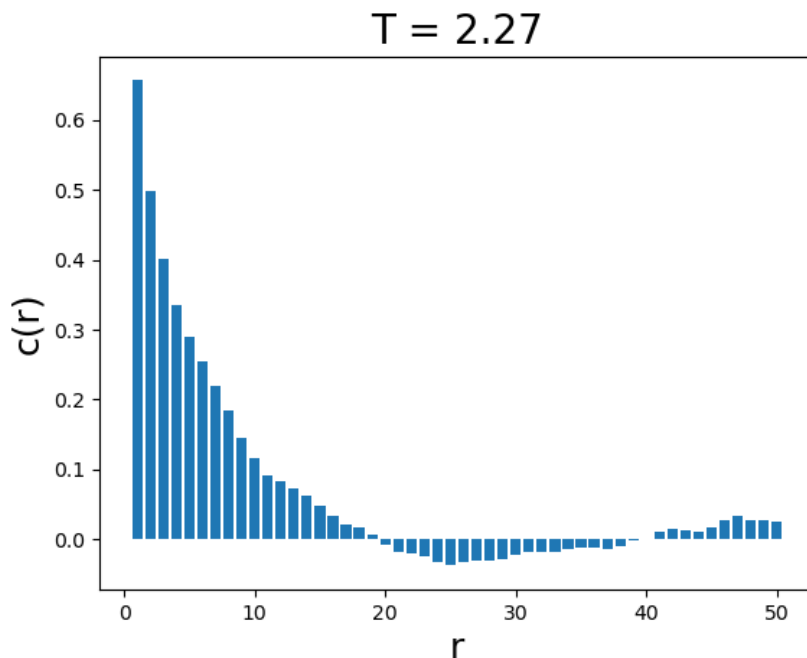
To quantify the clustering of alignments within an Ising magnet, we define a quantity called the **correlation function**, $c(r)$. Take any two dipoles, i and j , separated by a distance r , and compute the product of their states: $s_i s_j$. This product is 1 if the dipoles are parallel and -1 if the dipoles are antiparallel. Now average this quantity over *all* pairs that are separated by a fixed distance r , to obtain a measure of the tendency of dipoles to be "correlated" over this distance. Finally, to remove the effect of any overall magnetization of the system, subtract off the square of the average s . Written as an equation, then, the correlation function is

$$c(r) = \overline{s_i s_j} - \overline{s_i}^2$$

where it is understood that the first term averages over all pairs at the fixed distance r . Technically, the averages should also be taken over all possible states of the system, but don't do this yet.

(a) Add a routine to the `ising` program to compute the correlation function for the current state of the lattice, averaging over all pairs separated either vertically or horizontally (but not diagonally) by r units of distance, where r varies from 1 to half the lattice size. Have the program execute this routine periodically and plot the results as a bar graph.

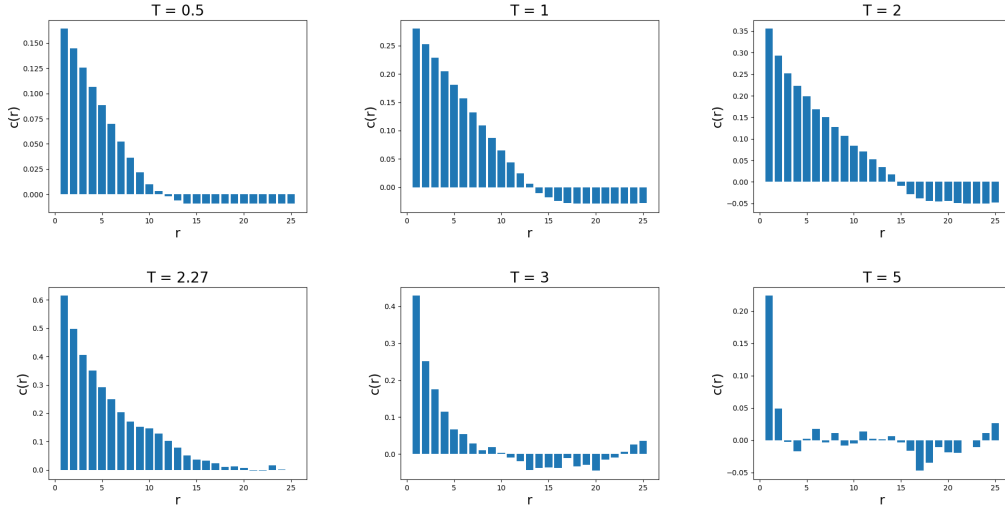
Answer: I ran the correlation for a lattice of size 100 and $T = 2.27$ and created the following bar graph:



We can see the function decays exponentially as r increases and after around $r = 20$ it is roughly zero.

(b) Run this program at a variety of temperatures above, below, and near the critical point. Use a lattice size of at least 20, preferably larger (especially near the critical point). Describe the behavior of the correlation function at each temperature.

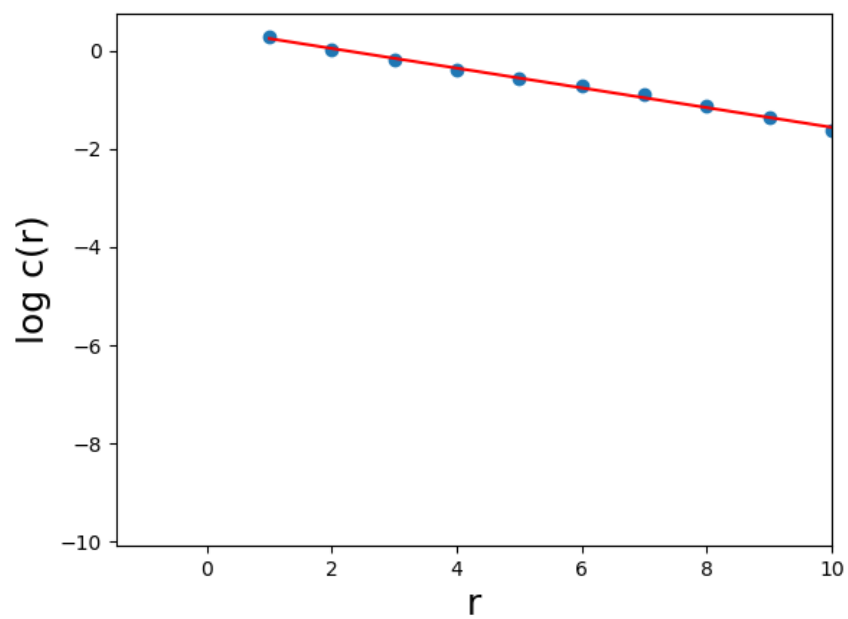
Answer: I created charts of the correlation for the following temperatures: 0.5, 1, 2, 2.27 (the critical temperature), 3, and 5, using a lattice size of 50 so that equilibrium can be reached quicker:



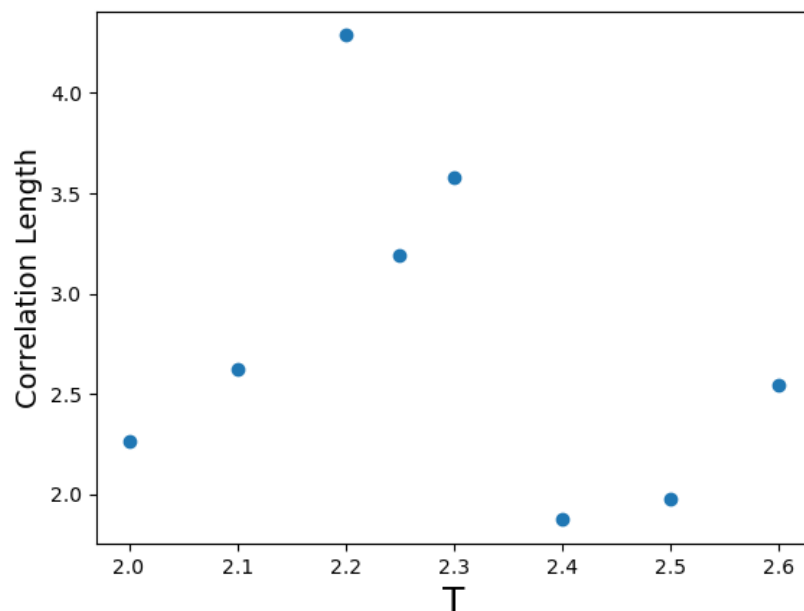
We can see that at temperatures below T_c , the correlation function is small in absolute terms, even at $r = 1$, but correlation continues to large distances. This makes sense because the lattice is rather homogeneous at equilibrium. Above T_c , the correlation function drops to 0 rapidly because there are at most small structures, and the magnetization as a whole is largely uncorrelated.

(c) Now add code to compute the *average* correlation function over the duration of a run. (However, it's best to let the system "equilibrate" to a typical state before you begin accumulating averages.) The **correlation length** as defined as the distance over which the correlation function decreases by a factor of e . Estimate the correlation length at each temperature, and plot a graph of the correlation length vs. T .

Answer: I opted to sample correlation functions from 50 of the last 1000 frames of the simulation and take the average of them. By converting to a scatter plot and log scale, I can come up with a linear fit to estimate the correlation length, which should be the negative reciprocal of the slope:



I was able to plot correlation lengths for a series of temperatures surrounding the critical temperature, with a lattice size of 30:



While this data is a bit noisy, it can be seen that the correlation length sharply increases around T_c . Above this temperature, large scale structures are broken up so dipoles are uncorrelated. Below this temperature, magnetization is unified at equilibrium, so there is no meaningful correlation to be ascertained. Only around the critical temperature do these effects balance out and yield a significant correlation length, signifying the unique structures found at T_c .

4 Problem 8.32

One of the most interesting features of the Ising model is the existence of a phase transition at a critical temperature T_c . Below the critical temperature the spins tend to align, producing a magnetized state, while above the critical temperature random fluctuations dominate and the spins become disordered. Near the critical temperature the system develops structures on many different scales, making the behavior difficult to analyze.

A powerful method for studying systems near critical points is the *renormalization group transform*. The central idea behind the renormalization group is to examine how a physical system changes when we “zoom out” and ignore small scale details. In the Ising model this can be accomplished using a procedure called a *block spin transformation*.

4.1 Block Spin Transformations

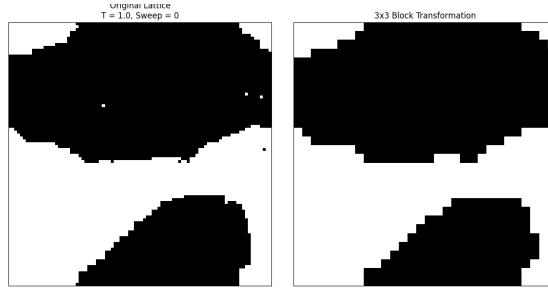


Figure 2: Example of a 3×3 block spin transformation applied to a 90×90 lattice at $T = 0$. Each 3×3 grid is replaced by a single site whose spin is the majority of the original block.

Instead of studying every microscopic spin individually, in a block spin transform, neighboring sites are grouped together into larger blocks and replaced by a single effective spin.

As example, we will divide the lattice into 3×3 blocks. Each block contains nine spins. The spins inside each block are replaced by a single new spin according to the *majority rule*:

$$S_{\text{new}} = \begin{cases} +1 & \text{if most spins point up,} \\ -1 & \text{if most spins point down.} \end{cases}$$

Since each block contains nine spins, ties are impossible.

Applying this transformation to the entire lattice reduces the size of the lattice by a factor of three in each direction. For example, a 90×90 lattice

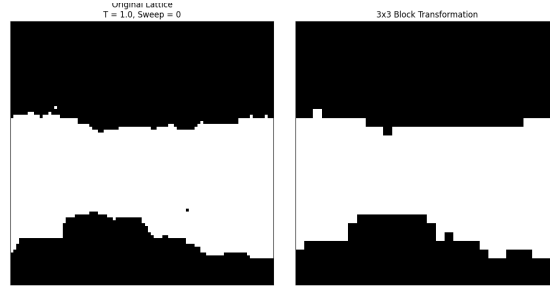
becomes a 30×30 lattice after one transformation as seen in Figure 2. Notice that after the block spin transformation, the transformed lattice resembles the structure of the original lattice, but with less precision.

As discussed earlier, the Ising model was simulated using the Metropolis algorithm. During each Monte Carlo sweep, a random spin was selected and flipped with probability

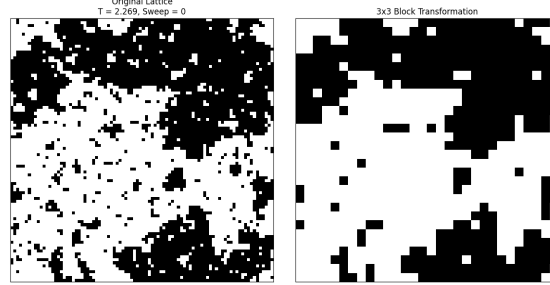
$$P = e^{-\Delta U/T},$$

whenever the energy increased by an amount $\Delta U > 0$. If the energy decreased, the flip was always accepted. After allowing the system to equilibrate, a block spin transformation was applied to the lattice. This was the process used to create Figure 2.

4.2 Equilibrating Systems



(a) Single lock transformation at a low temperature ($T = 1$) after the system has equilibrated.



(b) Single transformation near the critical temperature ($T \approx T_c$) after the system has equilibrated.



(c) Single transformation at a high temperature ($T = 5$) after the system has equilibrated.

To study the effects of this transformation, simulations were performed on a 90×90 lattice at several temperatures above, below, and near the critical

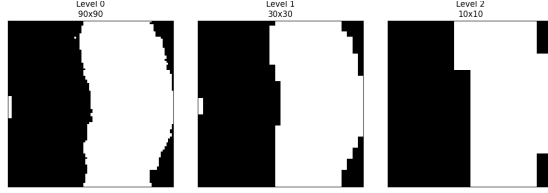
temperature.

At temperatures well below the critical temperature, such as $T = 1.0$, the lattice quickly evolved into a highly ordered state as seen in Figure 3a. Before the block spin transform was applied the system equilibrated into several large structures with similar spins, as predicted by the Ising model at low temperatures. After the block spin transformation was applied, the transformed lattice appeared even more ordered than the original lattice. Small isolated fluctuations disappeared during the averaging process. This behavior suggests that the transformed lattice corresponds to a lower effective temperature than the original system.

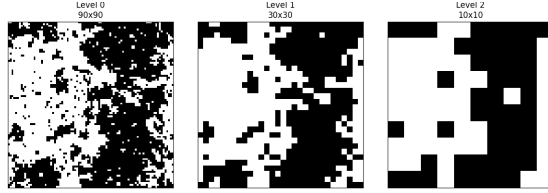
At temperatures well above the critical temperature, such as $T = 5.0$, the lattice appeared highly disordered as seen in Figure 3b. Before the transformation the system equilibrated into a disordered state with lots of small structures. The transformed lattice remained disordered after applying the transformation. In fact, the transformed lattice often appeared even more random because the majority rule averaged over already uncorrelated spins. This indicates that the effective temperature increases under the transformation when the original system is above the critical temperature.

Near the critical temperature, around $T_c \approx 2.269$, structures of many different scales formed. After the block spin transformation was applied, the transformed lattice looked very similar to the original lattice. This indicates that the transformation does not change the effective temperature for temperatures near the critical temperature.

4.3 Repeated Transformations



(a) Repeated block transformations at a low temperature ($T = 1$).



(b) Repeated block transformations near the critical temperature ($T \approx T_c$).



(c) Repeated block transformations at a high temperature ($T = 5$).

As seen earlier, applying the block spin transformation changes the effective temperature. Thus, if we apply the transformation several times to the same system, we should expect to see the effective temperatures converge to one of several fixed points.

If the initial temperature is less than the critical temperature then repeated transformations cause the system to become increasingly ordered. Structures of aligned spins grow larger while fluctuations become less important. Eventually, after enough transformations the effective temperature approaches $T = 0$. This can be seen in Figure 4a where after each block transformation the system looks like a system at a lower temperature, rapidly approaching zero effective temperature.

If the initial temperature is greater than the critical temperature then repeated transformations cause the system to become increasingly disordered. Correlations between neighboring spins disappear and the system approaches a completely random state. Eventually, after enough transformations the effective temperature approaches $T = \infty$. This can be seen in Figure 4c where after each block transformation the system looks like a system at a higher temperature, rapidly approaching infinite temperature.

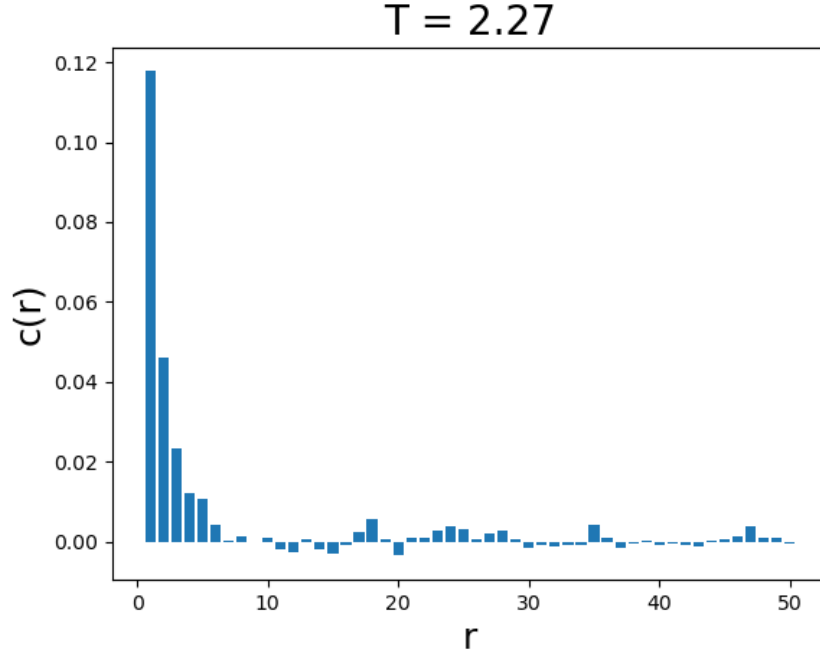
At the critical temperature, $T = T_c$, the system is approximately the same under repeated transformations. This occurs because near the critical temperature the block spin transformation does not qualitatively change the structures in the system. Because the system appears similar at different scales, the critical temperature is itself a fixed point of the renormalization group transformation. This can be seen in Figure 4b where after each block transformation the system looks to be at roughly the same temperature.

5 External Applied Field

If an external magnetic field is applied to our lattice, it tacks on another term to our energy:

$$U = -\epsilon \sum_{i,j \text{ pairs}} s_i s_j - B \sum_i s_i$$

In the Metropolis algorithm, this makes the state aligned with the magnetic field much more likely. This is reflected in the correlation function among other things. Consider the example of $T = 2.27$ now with a magnetic field $B = 0.05$:



In this case the lattice converged almost completely to a unified magnetization, which is reflected in the correlation function being 0 except for small scale structures.

6 Conclusion

While the Ising model is not a perfect representation of the quantum mechanics in ferromagnets, it is impressive that the model predicts a phase transition at the critical temperature, just as real permanent magnets will break down above their Curie temperature. This T_c can be identified from the correlation function and from visual inspection of self-similarity. Manipulating the model through a block spin transformation can alter behavior above and below T_c , but the critical temperature is ultimately preserved.